Band-1 Hadamard Order 7396 — Miyamoto (Paley-I DIRECT H_1848 + C_1850 GF(43^2)) + Paley-II IN SPK; Gap 2052 Still OPEN

H_1848 Paley-I PASS; C_1850 PASS; H_7396 Miyamoto PASS; H_7396 Paley-II PASS → IN SPK via Paley-II q=3697; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_7156_Miyamoto`; `2026-09-15_Band1_Hadamard_6724_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_7396_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 1848 and 7396 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Paley-I DIRECT seed H_{1848} = Paley-I($q = 1847$) and symmetric Paley conference C_{1850} over GF(43^2) ($q = 1849 \equiv 1 \pmod 4$), **and** classical Paley-II H_{7396} = Paley-II($q = 3697$). Classical SPK for 7396: Paley-I **FAIL**s ($q = 7395 = 3 \cdot 5 \cdot 17 \cdot 29$ composite); Paley-II **PASS**es ($q = 3697$ prime $\equiv 1 \pmod 4$); Kronecker **FAIL**s (4×1849). Therefore order **7396 is IN SPK via Paley-II** — same honesty class as 6724 / 6436 (Miyamoto + IN SPK). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **7156 Miyamoto-only (NOT in SPK)**. Prior SPK seeds **1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / 1668 / 6628 / 1656 / 1620 / 1608 / 6436 / 212 / 424 / 1696** retained; Miyamoto-only bonuses **7156/7108/6836/6788/6676/6484/6404/6388/6196/...** excluded; **1848** and **7396** folded into ONBOX/SPK when verified.

Abstract

Finite constructions of orders **1848** and **7396**.

- H_{1848} = Paley-I ($q = 1847$) — DIRECT (NOT Double)
- C_{1850} = Paley conference over GF(43^2), $q = 1849$
- H_{7396} = Miyamoto(H_{1848}, C_{1850})
- H_{7396} = Paley-II ($q = 3697$) — classical SPK path
- Paley-I for 7396: **SKIP** ($q = 7395$ composite)
- Kronecker for 7396: **SKIP** ($4 \cdot 1849$ not admissible)

SPK: 7396 IN SPK via Paley-II $q = 3697$. Seeds 1848 and 7396 enter SPK. Prior SPK **1788 / 888 / 1776 / 860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_7396.py`

Results. overall_pass=true; units 113/113; wall 656.7095s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 7396 = 4 \cdot 1849$; $1849 = 43^2 \equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1848} and C_{1850} (Paley conference over GF(43^2)).
- Seed: Paley-I $q = 1847$ (prime $\equiv 3 \pmod 4$) → H_{1848} **DIRECT** (NOT Double; Paley-II/Double half FAIL).
- SPK: Paley-I $q = 7395$ **FAIL** (composite $3 \cdot 5 \cdot 17 \cdot 29$); Paley-II $q = 3697$ **PASS** (prime $\equiv 1 \pmod 4$); Kronecker **FAIL → IN SPK via Paley-II**.
- Prior: 7156 Miyamoto-only shipped; skip 7204; this ship $q = 1849 \rightarrow n = 7396$.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_1848 Paley-I	Paley-I q=1847 DIRECT	0	4.6812
C_1850	Paley conf GF(43²) q=1849	0	3.648
H_7396 Miyamoto	Miyamoto(H_1848 Paley-I DIRECT, C_1850)	0	316.8153
H_7396 Paley-II	Paley-II q=3697 (classical SPK)	0	323.3282

Object	Method	max|HH^T-nI|	Wall (s)
H_7396 Paley-I	SKIP	—	—
	q=7395=3 · 5 · 17 · 29		
H_7396	SKIP 4 · 1849	—	—
Kronecker	not admissible		

SPK closure note

Seeds include prior SPK ships through **7156** lineage (**1788** Paley-I DIRECT; **888/1776**; **860/1720/6884**; **1708** Paley-II; **6724** Paley-II; **840/1680**; **6628**; **1668**; **1656**; **1620**; **1608**; **6436**; **212/424/1696**; **NOT** 7156/7108/6836/6788/6676/6484/6404/6388/6196) + verified **1848** (Paley-I DIRECT) + verified **7396** (Paley-II); **exclude** Miyamoto-only bonuses. Known count 1455. Next constructive gap **2052** (OPEN).

Quantity	Value
7396 in SPK	**True** (Paley-II q=3697)
1848 in SPK	**True** (Paley-I q=1847 DIRECT)
1788 in SPK	**True** (prior 7156 seed)
7156 / 7108 / 6836 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture · no Clay-closed · no finished-ToE · no RH / P!=NP / YM / Frankl / NS-solved / AC-solved · gap 2052 remains OPEN · finite constructions of 1848+7396 ONLY.

Repro

Lab: `/workspace/experiments/band1/hadamard_7396_miyamoto/`

Artifacts: `H1848_paleyI_pm1.npz`, `C1850_pm1.npz`, `H7396_miyamoto_pm1.npz`, `H7396_paleyII_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.